

Exp.No -5 Data Collection and Initial Exploration

PROGRAM:

```
import pandas as pd

dataset= {

    "Name": ["Arun", "Bala", "Charan", "Divya", "Esha"],

    "Age": [20, 21, 20, 19, 21],

    "Marks": [85, 78, 90, 88, 76]

}

df = pd.DataFrame(dataset)

df.to_csv("data.csv", index=False)

print("data.csv created successfully")

data = pd.read_csv("data.csv")

print("First 5 rows:")

print(data.head())

print("\nLast 5 rows:")

print(data.tail())

print("\nShape:")

print(data.shape)

print("\nColumn Names:")

print(data.columns)

print("\nInformation:")

data.info()

print("\nStatistical Summary:")

print(data.describe())

print("\nMissing Values:")

print(data.isnull().sum())
```

OUTPUT:

data.csv created successfully

First 5 rows:

	Name	Age	Marks
0	Arun	20	85
1	Bala	21	78
2	Charan	20	90
3	Divya	19	88
4	Esha	21	76

Last 5 rows:

	Name	Age	Marks
0	Arun	20	85
1	Bala	21	78
2	Charan	20	90
3	Divya	19	88
4	Esha	21	76

Shape:

(5, 3)

Column Names:

Index(['Name', 'Age', 'Marks'], dtype='str')

Information:

<class 'pandas.DataFrame'>

RangeIndex: 5 entries, 0 to 4

Data columns (total 3 columns):

#	Column	Non-Null Count	Dtype
0	Name	5 non-null	str
1	Age	5 non-null	int64
2	Marks	5 non-null	int64

dtypes: int64(2), str(1)

memory usage: 252.0 bytes

Statistical Summary:

	Age	Marks
count	5.00000	5.00000
mean	20.20000	83.40000
std	0.83666	6.14817
min	19.00000	76.00000
25%	20.00000	78.00000
50%	20.00000	85.00000
75%	21.00000	88.00000
max	21.00000	90.00000

Missing Values:

Name 0

Age 0

Marks 0

dtype: int64